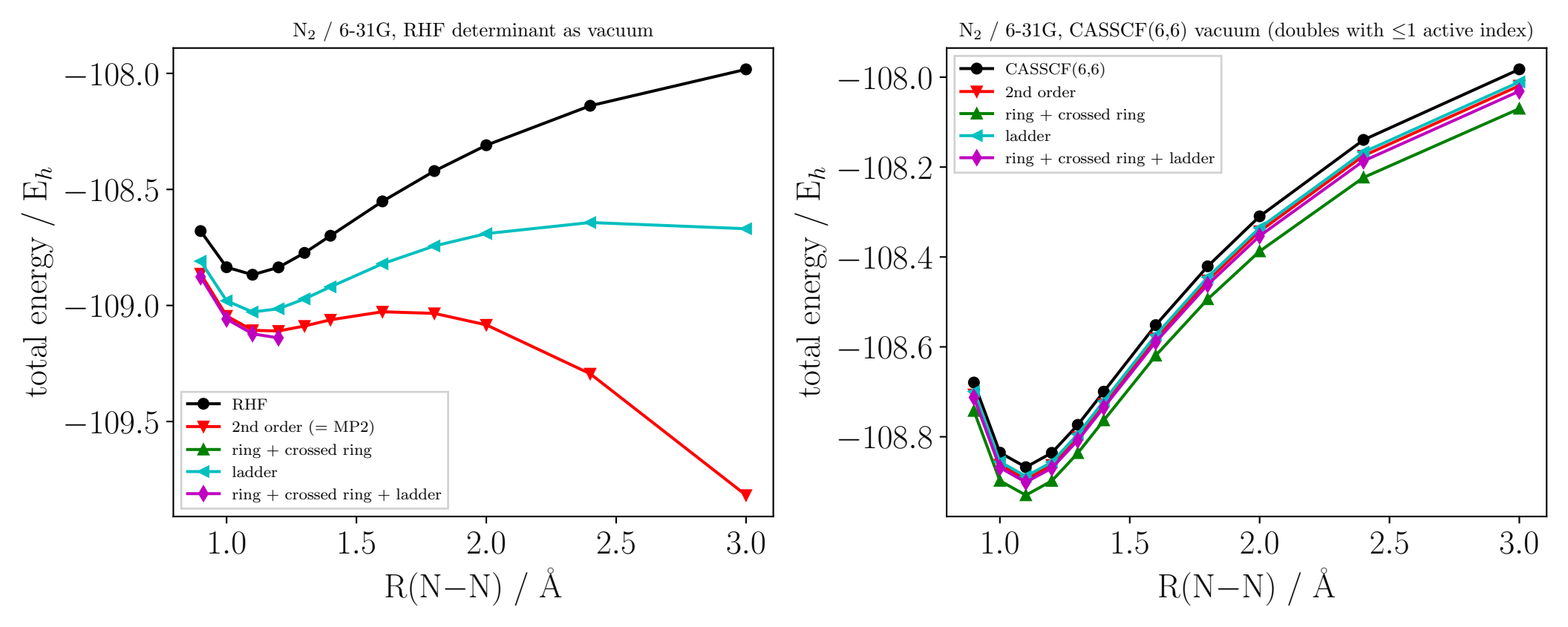

N2 / 6-31G, RHF determinant as vacuum
total energy / Eh
R(N−N) / Å
−108.0
−108.5
−109.0
−109.5
1.0
1.5
2.0
2.5
3.0
RHF
2nd order (= MP2)
ring + crossed ring
ladder
ring + crossed ring + ladder
N2 / 6-31G, CASSCF(6,6) vacuum (doubles with ≤1 active index)
total energy / Eh
R(N−N) / Å
−108.0
−108.2
−108.4
−108.6
−108.8
1.0
1.5
2.0
2.5
3.0
CASSCF(6,6)
2nd order
ring + crossed ring
ladder
ring + crossed ring + ladder